Agroforestry and Landscape Biodiversity

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Agroforestry and Landscape Biodiversity (A32365)

Short Title:	Agroforestry and Biodiversity		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Agriculture		
Author	JGERAGHTY		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

The module will provide participants with an in depth appreciation for the role of agroforestry and biodiversity as a vital part of a functional landscape approach in organic and biological production systems. Participants will be able to identify native species and multi-purpose uses of trees from food, fodder, and fuel production to habitat provision. The vital role and management of field boundary hedgerows for biodiversity, carbon sequestration and water cycling will be emphasised as will the importance of developing linked habitats for both domestic and wild native species. Participants will develop field survey skills in aquatic and terrestrial habitats for biodiversity assessment, evaluation, and monitoring. Support initiatives and policy developments will be reviewed in the context of their relevance to incorporating agroforestry and biodiversity management for ecosystem services and climate resilient farming systems.

Programmes

`SCIE-0084`	MSc in Organic and Biological Agriculture (`WD_SOBIA_R`)	2 / 3 / M
`SCIE-0084`	Postgraduate Programmes in Science (`WD_SGRAD_XX`)	5 / 0 / E

Indicative Content

- Agroforestry systems forming integrated functional landscapes in regenerative agriculture
- Silvo-pasture and silvo-arable systems for enhanced productivity and profitability
- Right tree in the right place to deliver integrated systems and ecosystem services
- Hedgerow management for enhanced soil carbon, biodiverse habitats, and livestock protection
- Role of landscape biodiversity for carbon and water cycling in climate resilient ecosystems
- Farming to conserve and enhance functional habitats for both domestic and wild species
- Field surveying for assessing, evaluating, and scoring biodiverse landscapes
- Policy support and regulations for agroforestry and biodiverse habitat provision under Pillar 1 and Pillar 2 of the CAP

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe and explain the beneficial role of and ecosystem functions provided by agroforestry and biodiverse landscapes.
2. Review, recognize and compare different silvo-pasture and silvo-arable methods and systems being practised across Europe.
3. Identify and select native tree species with suitable growth habits in different environments for their satisfactory development.
4. Devise, design and develop an integrated agroforestry plan for a selected farm landscape and crop or livestock production system.
5. Assess and evaluate the quality of, and produce remedial actions for, the enhancement of field boundaries and hedgerow habitats on a sample farm..
6. Conduct a biodiversity analysis and survey of a farm landscape and prepare and describe a biodiversity enhancement plan that sensitively integrates domestic and native wild species.
7. Analyse, interpret and summarize research findings and developing support policy for agroforestry and biodiversity and their likely impact for developing climate resilient and functional farming landscapes.

Learning and Teaching Methods

- Online lectures, guest lectures and webinars
- Assignment for class discussion and self-directed learning
- Case studies on agroforestry and enhanced biodiversity on organic farms

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5,6,7
Case Studies	50%	1,2,3,4,5,6,7
Case Studies	50%	1,2,3,4,5,6,7

Assessment Criteria

<40%: Very limited knowledge and understanding of the indicative content and topics covered. Unable to carry out critical thinking or practical skills.

40%--59%: Will be able to show general knowledge of the key learning outcomes, including an appreciation for the contribution that agro-forestry and biodiversity can make to the agricultural landscapes. The student will be able to write and present a quality development plan for agro-forestry and habitat for biodiversity.

60%--69%: In addition to the above, the student will be able to demonstrate an in-depth understanding of the inter-relationships between agro-forestry and biodiversity in the landscape and identify how these may be used to improve farm productivity. The student be able to write and present a good quality plan for agro-forestry and biodiverse habitat development.

70%--100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the vital role of agro-forestry and biodiversity in climate resilient farm enterprises and create a high quality plan for agro-forestry and biodiverse habitat development for an existing farm.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial		12
Practical		12
Online Delivery		24
Independent Learning		222

Essential Material

- "National Biodiversity Data Centre." www.biodiversityireland.ie/resources
- "Woodland Improvement Scheme." https://www.gov.ie/en/service/d54212-woodland-improvement-scheme/

Supplementary Material

- "Agroforestry Conference Proceedings." 12-13th April 2021. https://nots.ie/events/agroforestry-conference-2021-april-12th-13th/
- Vero, S.E. _Fieldwork Ready - An Introductory Guide to Field Research_. "1st ed.". USA: John Wiley & Sons, 2021.